

Markscheme

Mathematics: analysis and approaches

Practice question bank

101. (a)

It can be proven by contradiction that there is no smallest positive rational number (assume one exists, then halve it to find a smaller one, a contradiction).

Using this method, find a positive rational number smaller than $x = 0.0001$, of the form $\frac{x}{k}$ with $k = 2$.

$$\frac{x}{2} = \frac{0.0001}{2} \quad (M1)$$

$$= 0.00005 \quad \text{A1}$$

[3 marks]